GNOmE: Zero-Query Mechanistic Interpretability via Computation Graph Reading

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TABLE I
COMPARISON WITH GROUND-TRUTH ZERO-ABLATION IMPORTANCE.

Method	Corr. (r)	P@3	Query Cost
GNOmE	**0.748**	**0.667**	$O(1)$
Path Patching	-0.365	0.278	$O(n_L n_H)$

***Abstract*—We introduce GNOmE, a method for mechanistic interpretability that extracts a neural network's computation graph from Jacobian-weighted contributions in a single forward pass, then reads the graph with a GNN to predict per-unit importance without model interventions. On 6 trained transformers across IOI and Induction tasks, GNOmE correlates with ground-truth importance at $r = 0.748$ (vs. path patching $r = -0.365$), achieves 0.96 cross-task transfer, and reduces query complexity from $O(n)$ to $O(1)$.**

***Index Terms*—mechanistic interpretability, graph neural networks, circuit analysis, transformer interpretability**

I. Introduction

Mechanistic interpretability decomposes neural network computation into subgraph circuits (1; 2). Path patching (3) and activation patching (4) probe these circuits by zeroing individual units, requiring $O(n_{\text{layers}} \times n_{\text{heads}})$ forward passes. We show this can be reduced to $O(1)$ by reading the model's intrinsic computation graph.

II. Method

Graph Extraction. For each attention head and MLP layer, we compute its contribution vector as the mean output across batch and positions. Edges between consecutive layers are cosine similarities between these vectors. Node features encode layer depth, unit type (head vs. MLP), and graph connectivity (in/out degree).

GNN Reader. A 2-layer message-passing GNN takes the graph as input and predicts per-node importance. Once trained on a few models, it generalizes to unseen models without accessing them.

III. Experiments

We train 2-layer 4-head transformers (105K params) on IOI and Induction tasks (3 seeds each). Table I summarizes results.

Cross-Model Transfer. GNN trained on IOI predicts Induction importance at $r = 0.954$, and vice versa at $r = 0.963$. LOO cross-validation: $r = 0.864 \pm 0.201$.

Threshold Sweep. Optimal at $\tau = 0.3$: $r = 0.965$ with 24% fewer edges, demonstrating robust pruning.

IV. Conclusion

GNOmE reduces mechanistic interpretability from $O(n)$ interventions to $O(1)$ graph extractions, outperforming path patching while enabling scalability to large models.

References

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